Summary of the Project

This project focuses on understanding whether the existing ML and NLP workflows that I had already created could behave like reusable analytical systems instead of remaining static project structures. The main objective was to understand whether future market data and future text data could continuously enter the same workflow with minimal rebuilding, controlled manual effort, and stable downstream continuity.

The project mainly worked across two different operational architectures. The ML workflow behaved like a centralized dependency-connected structure because BigQuery ML directly supports machine learning functions inside the environment itself. In contrast, the NLP workflow behaved much more modularly because NLP extraction functions required external Python-based processing before reconnecting the outputs back into BigQuery SQL for downstream analysis continuity.

In the ML workflow, the project focused on building reusable raw-input continuity, append continuity, and downstream synchronization so future market data could enter the same inference structure without rebuilding the workflow again. The implementation successfully showed that the ML architecture can behave like a reusable inference-based system when schema continuity and operational compatibility remain stable.

In the NLP workflow, the project focused on building reusable extraction continuity for signals such as sentiment, entities, phrases, and earnings sentiment. One of the biggest improvements achieved was creating a reusable framework for future news extraction using Google News RSS and Python libraries, which significantly improved the continuity of future news-data collection.

The project also helped highlight that reusable systems are not fully automatic in nature. While reusable extraction continuity and downstream operational continuity can survive successfully, some manual analytical continuity still remains necessary, especially in the NLP theme-analysis workflow where interpretation and refinement continue evolving with new text data.

Overall, the project demonstrates that both the ML and NLP workflows can evolve beyond static analytical structures and behave more like reusable operational systems where future data can continuously enter the same architecture with stable schema continuity, reusable extraction continuity, and manageable downstream operational stability.

Key Findings

One of the biggest findings of this project is that reusable analytical systems mainly survive through stable schema continuity rather than through fixed data itself. Future market data and future text data can continuously change over time, but if the schema structure, datatypes, naming continuity, and downstream dependencies remain stable, then the reusable workflow continuity can also remain operationally stable.

Another major finding is that the ML and NLP workflows survive reusable continuity in very different architectural ways. The ML workflow behaves like a centralized dependency-connected system where downstream continuity remains highly interconnected. Because of this, even small schema inconsistencies in the raw-input layer can propagate throughout the downstream workflow. In contrast, the NLP workflow behaves much more modularly, where every signal continuity survives more independently through reusable extraction layers and separate operational validation.

The project also showed that BigQuery ML creates a much smoother reusable operational structure compared to NLP workflows because machine learning functions are directly supported inside the environment itself. Meanwhile, NLP workflows become operationally more complex because the extraction continuity depends heavily on external Python-based processing before reconnecting with the downstream SQL structure.

Another important finding is that reusable systems are not fully automatic in nature. Even when reusable extraction continuity and downstream synchronization survive successfully, some level of manual operational continuity still remains necessary. This became especially visible in the NLP theme-analysis workflow where new phrases and evolving themes continuously require interpretative refinement and analytical restructuring.

The project also highlighted the importance of reusable ingestion frameworks. Through reusable Python-based extraction structures, merge workflows, and append continuity frameworks, future data continuity became much easier to integrate operationally without rebuilding the entire workflow repeatedly. This became one of the strongest operational improvements achieved across both systems.

Overall, the project demonstrated that reusable analytical continuity is possible in both ML and NLP workflows when schema continuity, operational compatibility, reusable extraction continuity, and downstream synchronization are preserved properly across the overall architecture.

What Worked

One of the biggest things that worked successfully in this project was the reusable continuity of the ML workflow. After building the reusable extraction framework, append continuity, and schema-preserving ingestion structure, the new market data was successfully integrated into the existing ML architecture without retraining the models or rebuilding the downstream workflow. The downstream views, inference continuity, and analytical structures updated successfully after the appended data entered the system.

Another major thing that worked successfully was the reusable extraction continuity built for the NLP news workflow. The Google News RSS framework and Python-based extraction process significantly improved the continuity and availability of future news-data collection compared to the original NLP workflow, where historical news scarcity had become a major limitation.

The reusable Python extraction templates for sentiment extraction, entity extraction, phrase extraction, and earnings sentiment extraction also remained operationally compatible with the downstream BigQuery SQL workflow. By preserving the same VADER and spaCy-based extraction continuity that existed in the original NLP workflow, the regenerated outputs successfully synchronized with the downstream analytical structure without major schema or compatibility failures.

The project also successfully demonstrated that SQL views can behave like reusable operational continuity layers instead of static analytical outputs. Once the schema continuity remained stable, the downstream workflow continuity updated automatically when the new appended data entered the system. This became one of the strongest operational validations of the reusable architecture itself.

Another important thing that worked successfully was the modular checkpoint-based structure created through the Scrum approach. Dividing the workflows into smaller operational checkpoints, validation layers, and signal-based structures made the reusable system much easier to analyse, validate, and debug operationally whenever new continuity entered the workflow.

Overall, the project successfully proved that both the ML and NLP workflows can survive future appended-data continuity operationally when schema continuity, extraction continuity, and downstream compatibility continuity are preserved properly across the reusable architecture.

What Did Not Work

One of the biggest limitations that still remains in the project is that the NLP workflow is not fully reusable in the same smooth way as the ML workflow. Since BigQuery SQL does not directly support NLP extraction continuity, the NLP architecture still depends heavily on external Python-based processing and manual operational handling before reconnecting the regenerated outputs back into the SQL workflow.

Another major limitation is the earnings-data extraction continuity. Unlike the market-return continuity and the news-continuity framework that were successfully made reusable, the earnings workflow still does not have a stable reusable extraction framework for collecting future earnings continuity automatically. Because of this, the earnings layer still remains partially manual operationally.

The project also highlighted that the theme-analysis workflow inside the NLP architecture cannot become fully automatic operationally. Even when the phrase extraction continuity survives correctly, new appended text continuity naturally introduces new phrases and evolving thematic structures. Because of this, manual analytical refinement and interpretative grouping of themes still remain necessary whenever future text continuity enters the workflow.

Another limitation that became visible is the operational fragility of replacing regenerated extraction files inside the NLP workflow. Since the downstream SQL continuity depends heavily on existing naming structures and signal continuity, deleting and replacing extraction files manually still remains tedious and error-prone operationally.

The project also revealed that the ML workflow, although operationally smoother, is highly sensitive to schema mismatches because of its centralized dependency-connected continuity. Even small inconsistencies in the raw-input continuity can propagate aggressively throughout the downstream workflow and create operational instability across multiple connected structures.

Overall, the project showed that while reusable analytical continuity is operationally possible across both the ML and NLP workflows, complete automation is still not realistically achievable in every layer. Manual operational supervision, schema validation, interpretative continuity, and reusable framework maintenance still remain important parts of sustaining the overall architecture.

Final Takeaway

The biggest takeaway from this project is that reusable analytical systems are not created by removing all manual work, but by creating stable operational structures where future compatible data can continuously enter the same workflow without rebuilding the complete architecture repeatedly.

Through this project, I understood that schema continuity is the real backbone of reusable continuity. If the schema structure, datatypes, naming continuity, and downstream dependencies remain stable, then the reusable analytical workflow can also remain operationally stable even when the future data itself continuously changes over time.

Another major understanding is that different systems survive reusable continuity differently. The ML workflow survives mainly through centralized dependency-connected continuity, while the NLP workflow survives mainly through modular reusable extraction continuity. Even though both architectures behave very differently operationally, both can still function together inside one broader reusable analytical ecosystem.

The project also showed that reusable systems are not fully automatic in nature, especially in NLP workflows where interpretative continuity and manual analytical refinement still remain necessary. Because of this, the goal of reusable architecture is not complete automation, but controlled operational continuity with manageable manual intervention.

Overall, this project helped transform both the ML and NLP workflows from one-time analytical projects into reusable operational systems capable of handling future appended-data continuity with much greater structural stability, operational understanding, and long-term scalability.